

Project Statement

Student Expense Tracker

Problem Statement

Students make a lot of small payments every day for things like food, travel, stationery, books, snacks and shopping. It is difficult to keep track of all of them by remembering or writing in a notebook. Because of this, many students do not know how much they spent in a month or whether they are staying inside the budget they planned.

The Student Expense Tracker is a simple Python project that helps to record, organize, search and check these expenses in one place.

Scope

This project is only for personal expense management. With it, a user can add, view, search, filter and delete expenses. It can also find the highest expense, calculate the spending of the current month, set a monthly budget, show budget warnings and generate a simple expense report.

The data is stored on the user's own computer in a JSON file, so no online database or internet is needed.

Target Users

- College students
- School students
- Students living in hostels
- Anyone who wants a simple record of their personal expenses

High-Level Features

1. Add and save expenses.
2. Keep expenses in categories.
3. Search and filter expenses.
4. Delete selected expenses.
5. Calculate the monthly spending.
6. Set and check a monthly budget.
7. Show budget warnings.
8. Find the highest expense.
9. Generate an expense report.
10. Save and load the expense data using JSON.

Technical Approach

The project is divided into separate Python files so that each file has its own work:

- main.py starts the application.
- gui.py makes the Tkinter window.
- expense_manager.py handles the expense operations.
- budget.py does the budget checking and status.
- reports.py makes the reports and totals .
- storage.py saves and loads JSON file.

In this project I used the Python basics we learned so far: variables, data types, conditions, loops, functions, classes, lists, dictionaries, file handling, exception handling and GUI programming.

Expected Outcome

The application should give students an easy way to note down their expenses and understand their monthly spending, without needing any database or an external charting library.

Project Relevance

This project uses Python programming concepts to solve a real problem that students face in daily life. Like, helps users set a budget and keep track of how much they have spent and to understand where their money is going. It is made for the VITyarthi Build Your Own Project evaluation.